

INFO90002 S2 2026

Assignment 2 - SQL

Due: see LMS

Submission: Via the Canvas LMS

Weighting: 10% of your total assessment (Graded out of 200 marks)

Hospital Management System

Community Hospital needs a system to keep track of hospital operations which are depicted in Figure 1. Note, you can download the figure from LMS which would be another option to zoom in and examine the entities and their relationships.

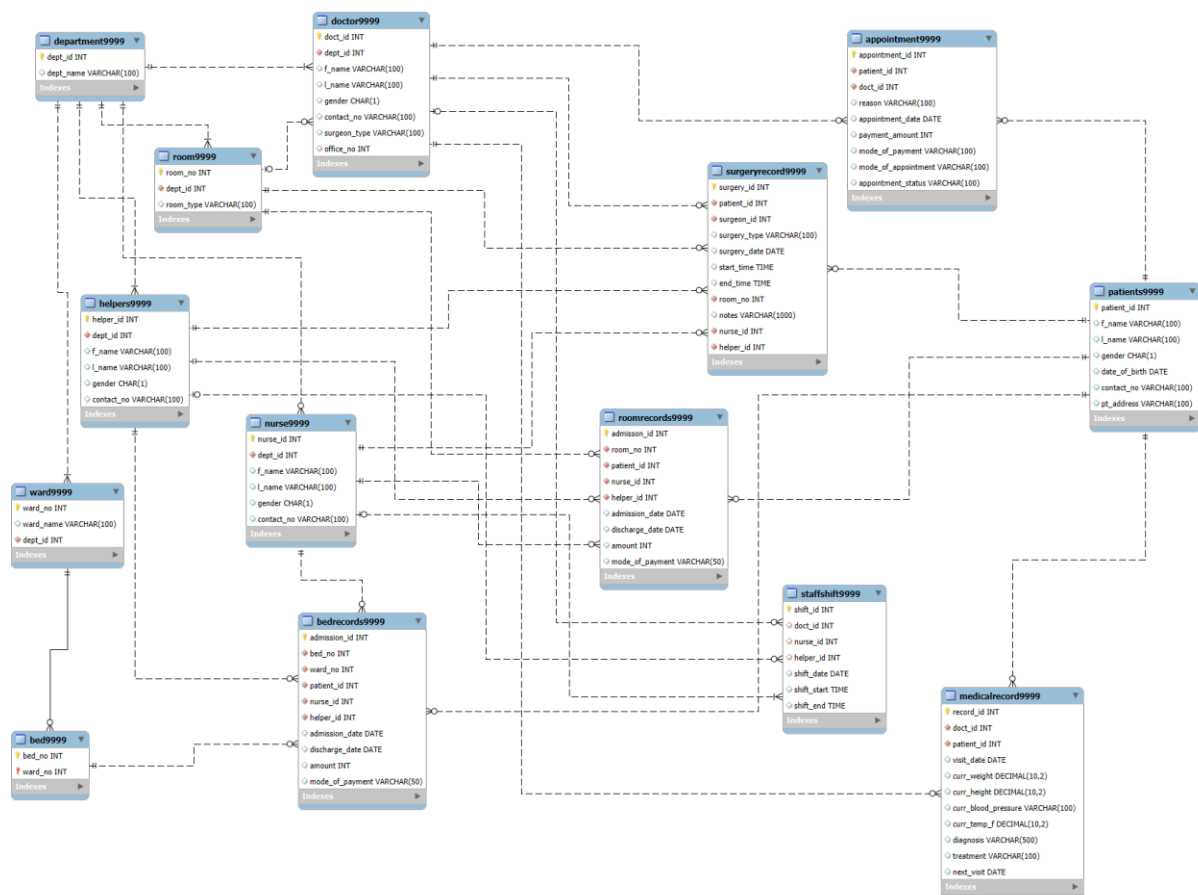


Figure 1. Data Model for Hospital Management System (the diagram is available for download from LMS)

The hospital beds are allocated to wards and wards belong to departments. Each department is in charge of multiple wards.

Each department has multiple doctors, nurses and helpers.

Patients will have their medical records stored in the database.

Patients that are not staying overnight in the hospital (i.e. no beds records for these patients, they are called outpatients) may have appointments with doctors.

Patients that need surgery will have surgery record containing the doctor (surgeon), surgery details, date and start / end time, room where the surgery was performed as well as associated nurse and helper.

Patients who need hospitalisation will have bed allocations.

Instructions

1. **Rename all tables** to have the last 4 digits in table names the same as the last 4 digits of your student ID
 - **Download** the files named **A2Script_2026s1_LOCAL.sql** and **A2Script_2026s1_ENG.sql** from the **LMS**.
 - Open each file in a text editor, e.g. in MySQL Workbench, Notepad, Notepad++ or some other text editor
 - Change all occurrences of **9999** to **the last 4 digits of your student ID** (one way to do this is to perform a **find and replace**). For example, if your student ID is 12349876, your tables will get renamed as room9876, bed9876, ward9876, etc.
Make sure you rename tables in BOTH sql files.
Note, if renaming is not done, you cannot get full marks, a heavy penalty of 50% will be applied.
2. Run the relevant script on the engineering server (and / or on your local MySQL server). This will create all required tables and populate them with data. Note on the local server you will have the database **hospitalmanagementsystem**, while on the engineering server you will only have a collection of tables (possibly mixed with tables from your previous exercises because students do not have permissions for *create database* and *create schema* commands execution).
3. Type the following SQL statement and execute it.

```
SELECT '123459999' as StuID, d.*  
FROM department9999 d
```

(Note that in the above statement 123459999 should be replaced with **your** Student ID and occurrences of 9999 should be replaced with the last 4 digits of your student ID, as in step 1 above)

Notice how each row contains your student id and all attributes from the department9999 table.

StuID	dept_Id	dept_Name
123459999	101	Cardiology
123459999	102	Neurology
123459999	103	Gastroenterology
123459999	104	Nephrology
123459999	105	Pulmonology
123459999	106	Endocrinology
123459999	107	General Medicine
123459999	108	Pediatrics
123459999	109	Orthopedics
123459999	110	Gynecology
123459999	111	Dentistry
123459999	112	Dermatology
123459999	113	Psychiatry
123459999	114	Psychology
123459999	115	Urology
123459999	116	ENT
123459999	117	Surgery
123459999	118	Radiology

You must include your student ID in all SELECT queries. 10% will be deducted from each task where this requirement is not satisfied.

It is expected that

- your script will produce correct results
- your code meets standards of quality as discussed in lectures
- your code runs on the university engineering server

The Task

For each question, write a single SQL statement. **Do not use** inline views / schema on read, views unless explicitly instructed to do so. Views, inline views and schema on read will **earn 0 marks**. Only question 11b is expected to use a view (not inline view).

If your result set is less than 10 rows, show ALL results. If it is longer, show at least 10 rows. Specify how many results were returned in red font under the screenshot.

Note the default setting in Workbench is 1000 records, even if there are many more in a table. So you may need to increase the limit, especially if you get exactly 1000 records returned.

Questions

1. How many surgeries did each patient have? Your results should show patient id, patient names as first and last name separated by space, and number of surgeries ordered by the total number of surgeries in descending order. Note, patients who did not have surgeries must not be listed.

(15 marks)

2. We need a list of doctors, showing for each doctor the total amount of payments for the doctor's appointments with patients. The list should also include doctors who did not have any appointments. Your results should show doctor IDs, doctor names as first and last names separated by space, and total amounts, ordered by the totals from highest to lowest.

(15 marks)

3. Produce a report showing number of shifts only with doctors on duty (no nurses or helpers), number of shifts with only nurses on duty (no doctors or helpers) and number of shifts with only helpers on duty (no doctors or nurses). Your report should use the following format and headings (actual staff numbers are removed):

StuID	Staff on duty	Total shifts
123459999	doctor shift	114
123459999	nurse shift	51
123459999	helper shift	721

(15 marks)

4. List all patients who have more than one cancelled appointment. You list should show patient id, last name and first name, gender and how many appointments this patient cancelled. Order the results by patients last name.

(20 marks)

5. List all helpers who have not been allocated to any staff shifts. Your list should show department id and name where the helper works, helper id and helper full name as first and last names separated by space. The results should be sorted by department id and within each department alphabetically by helper full name.

(20 marks)

6. How much money came through as payments in total and in each calendar year using each mode of payment. Your results should show full total in the first row and then rows with details: calendar year, mode of payment and money received ordered by year and mode of payment (similar to below but showing total values in the last column). Your output should look similar to the one below.

StuID	CalendarYear	mode_of_payment	PaymentsTotal
123459999		Total	----
123459999	2024	Card	-----
123459999	2024	Cash	-----

(20 marks)

7. List the number of surgeries performed by each surgeon in each calendar year. Your list should display years when surgeries were performed, then surgeons' id and last name and the number of performed surgeries.

(15 marks)

8. For each letter of the alphabet – A, B, C etc. – display the number of patients whose last name begins with that letter. Order your output alphabetically. If a letter doesn't have any corresponding patients, you can simply omit it from the listing. Hint: you can use the `LEFT(string, n)` function to find the first letter of a string. For example, the first two lines of your output would look like this:

Letter	HowMany
A	241
B	71

(10 marks)

9. Which patient (or patients if more than one) has had the highest number of appointments? (That is, more appointments than all the other patients.) If there is a tie for first place, list all the patients who tied. Your results should show the patient(s) ID, first and last name in the format "last comma space first" (e.g. Lee, June), and the number of appointments they had.

Hint: you will need to use ALL (see additional explanation https://www.w3schools.com/sql/sql_all.asp).

(25 marks)

10. Note, for this question StuID only needs to be inserted in the SELECT statement.

- a. Add yourself as a patient. You do not need to provide your real DOB, phone and address, any fictitious data is acceptable, as long as it is saved in the table. Use your student ID as patient ID.

Provide a screenshot of the message generated by Workbench that a record was added.

(3 marks)

- b. Add an appointment for yourself with Dr. Sonia Ali on any date in October 2026; use any reason you want for the appointment. The appointment status should be as "scheduled" and in person without payment details. This task needs to be done as a simple transaction where you need to determine doctor's ID first and only after that you can save appointment details.

Note, to find out the last appointment number use `MAX(appointment_Id)`.

This task needs to be done as a transaction.

Provide a screenshot of the status of executing the transaction. Make sure you include all status rows relevant to the transaction.

(12 marks)

- c. Create an SQL statement showing your id and name as a patient and your appointment details including appointment id, doctor's id and full name (last and first separated by comma and space), date, mode of the appointment and status of the appointment.

(10 marks)

11.

- a. Write the SQL DDL to create a view that counts number of wards in each department. Your list should include department ID, department name and number of wards.

Do not include StuID in this view.

In addition to the code, you need to provide 2 screenshots

- the list of tables and views from the left pane of Workbench showing your created view and
- the results of running SELECT from your View (we recommend explicitly showing the SELECT statement used to create the View).

(10 marks)

- b. Using the View you created in Task 11a, list the departments with the highest number of wards. Your results need to display department id, department name and the corresponding number of wards.

You **must** include your student ID in task b.

(10 marks)

Submission Details:

Submit a single PDF showing your answers to all questions

Please **make sure that you actually submit** your file on LMS. After uploading the file, click on 'Submit Assignment' to actually submit your file. If you submit late because you failed to press the submit button and only noticed this after the deadline, your submission will be considered late just like any other late submission to maintain fairness for all students.

Specify your student name and ID at the top of your answer document.

Formatting requirements for your submission

For each question, present an answer in the following format:

- Show the question number and question in **black** text.
- Show your answer (the SQL statement) in **blue** text (**DO NOT use a screen shot**)

- Show a screenshot of the result from Workbench. If your result set is less than 10 rows, show ALL results. If it is longer, show at least 10 rows (few rows extra is allowed).
- Show how many rows were actually returned, in **red** text.
- Remember to include **your student ID (the only exception is q.11a)**.
- Show each query on a separate page.
- You must not use in-line views, schema on read, views for questions unless explicitly instructed to do so (Q11b only can use a view).
- We may be testing your code on the engineering server so make sure you install the database and test your answers on the “engineering” MySQL server.

Example:

Q.XX List all doctors from departments 103 and 105. Your list should doctor_Id, first name, last name and department ID, ordered by doctor ID within department.

```
SELECT '123459999' as StuID, doct_Id, FName, LName, dept_Id
FROM doctor9999
WHERE dept_Id = 103 OR dept_Id = 105
ORDER BY dept_Id, LName
```

StuID	doct_Id	FName	LName	dept_Id
123459999	1014	Dr. Asif	Abbas Naqvi	103
123459999	1015	Dr. Saba	Asad Ansari	103
123459999	1013	Dr. Saifullah	Mir	103
123459999	1021	Dr. Shahbaz	Ali	105
123459999	1023	Dr. Shazia	Fakhar	105
123459999	1020	Dr. Waqas	Javed	105
123459999	1022	Dr. Abida	Raza	105

7 rows

Note, the screenshot should contain only data, no extra space to the right (empty parts of rows) and no need to include the SQL in the screenshot.

IMPORTANT: ATTEMPT EVERY QUESTION!

Ensure your scripts run on the engineering MySQL server

Important aspects we consider during marking

- All tables have correct last 4 digits – otherwise deduction from the final mark is 50%
- Student ID appears in all SELECT queries as the first column except question on creating a view or if there is a question where the student's ID is added as part of the INSERT – deduction 10% per question without ID in col 1
- Scripts not tested on the engineering server – 10% deduction from the overall assignment mark
- If the output on the screenshot does not match the SQL script – the mark will be 0. For example, SQL has more or less columns than the screenshot; or the script is not running but there is an attempt to deceive the marker by providing a screenshot.
- Deductions will be made on missing
 - INNER / OUTER in JOIN
 - Aliases, e.g. if the column name contains a function - this example is a real challenge for readability `ROUND(AVG(DATEDIFF()))`
 - ORDER BY in queries expecting it
- Deductions will be made
 - In cases with hardcoded values, e.g. hardcoding a year will make the query useless in the future
 - When treating data types incorrectly (even if MySQL is forgiving, we are not). For example, marks will be deducted if YEAR is treated as a string.

Please refer to lectures on the expectations of optimised queries that are not wasting CPU time (e.g. avoid using subqueries if a form of JOIN can be used instead).